

A disproof of Conjecture 00000000504

Projective dimension of edge ideals of r -partite r -uniform hypergraphs is unbounded

1 The conjecture

Conjecture 00000000504 reads (English version): “*Definition: The length of a minimal free resolution (projective dimension pd). Conjecture: $\text{pd } R/I(G)$ has threshold $2r - 1$ for r -partite r -uniform hypergraphs (generalizing the Froberg-type theorem $\text{pd} \leq 3$ for bipartite graphs).*” The Chinese version says the same: the threshold (the Chinese word means threshold / critical value) of $\text{pd } R/I(G)$ for r -partite r -uniform hypergraphs is $2r - 1$, generalizing the theorem $\text{pd} \leq 3$ for bipartite graphs.

Readings. Here $R = k[x_v : v \in V]$ is the polynomial ring over a field k and $I(G)$ is the edge ideal. The parenthetical (“generalizing ... $\text{pd} \leq 3$ for bipartite graphs”, and $2 \cdot 2 - 1 = 3$) forces the upper-bound sense of “threshold”. We refute:

- (A) $\text{pd}_R R/I(G) \leq 2r - 1$ for every r -partite r -uniform hypergraph G (for $r = 2$ this is the quoted bound $\text{pd} \leq 3$ for bipartite graphs);
- (B) the supremum of $\text{pd}_R R/I(G)$ over r -partite r -uniform hypergraphs equals $2r - 1$.

Both fail for every field k and every $r \geq 1$. **Not addressed:** a reading of the form “ $\text{pd} \leq 2r - 1$ if and only if G has some (unspecified) property”, or a reading in which “threshold” refers to an invariant other than pd ; the text does not determine such a reading. A lower-bound reading ($\text{pd} \geq 2r - 1$ for all such G) is not natural given the parenthetical and is not formalized.

2 Definitions (as formalized)

Definition 1. A *hypergraph* on a vertex type V is a finite set E of edges, each a finite subset of V . It is *r -uniform* if every edge has exactly r vertices. It is *r -partite* if there is a map $\text{part} : V \rightarrow \{0, \dots, r - 1\}$ (the parts are the fibres $V_j = \text{part}^{-1}(j)$) such that every edge meets every part V_j in exactly one vertex.

Definition 2. For a field k and $R = k[x_v : v \in V]$ (V finite), the *edge ideal* is $I(G) = (\prod_{v \in e} x_v : e \in E) \subseteq R$.

Projective dimension. $\text{pd}_R M$ is Mathlib’s `CategoryTheory.projectiveDimension` of M in the category of R -modules: the least n such that $\text{Ext}_R^i(M, -) = 0$ for all $i > n$, with values in $\mathbb{N} \cup \{\infty\}$ (and $\perp$ for $M = 0$). We only prove *lower* bounds on pd . Any free resolution of M of length ℓ shows $\text{pd}_R M \leq \ell$, so the length of a minimal free resolution is at least $\text{pd}_R M$; hence every lower bound below is also a lower bound for the length of the minimal free resolution named in the conjecture.

3 The counterexample

For $m \geq 0$ and $r \geq 1$ let $H_{r,m}$ be the r -uniform matching with m edges: vertex set $\{0, \dots, m-1\} \times \{0, \dots, r-1\}$ and edges $e_i = \{(i, 0), \dots, (i, r-1)\}$, $i < m$. Distinct edges are disjoint; $H_{r,m}$ is r -uniform, and r -partite with parts $V_j = \{(i, j) : i < m\}$ (each e_i meets V_j exactly in (i, j)). Its edge ideal is generated by the monomials $f_i = \prod_{j < r} x_{(i,j)}$, which involve pairwise disjoint sets of variables.

Lemma 3 (colon step). *Let J be the ideal generated by monomials x^d , $d \in S$, and let x^e be a monomial such that for every $d \in S$ and every variable v , $d_v = 0$ or $e_v = 0$. If $x^e g \in J$ then $g \in J$.*

Proof. A polynomial lies in a monomial ideal iff each of its monomials is divisible by a generator. If x^a occurs in g , then x^{a+e} occurs in $x^e g$, so $d \leq a + e$ for some $d \in S$. At a variable v with $e_v = 0$ this gives $d_v \leq a_v$; at a variable with $d_v = 0$ it is trivial. So $d \leq a$ and every monomial of g is divisible by a generator. $\square$

Lemma 4. *If $d_0, \dots, d_{m-1}$ are exponent vectors with pairwise disjoint supports, then $x^{d_0}, \dots, x^{d_{m-1}}$ is a weakly regular sequence in R : each x^{d_i} is a nonzerodivisor on $R/(x^{d_0}, \dots, x^{d_{i-1}})$.*

Proof. Lemma 3 with $S = \{d_0, \dots, d_{i-1}\}$ and $e = d_i$. $\square$

Lemma 5. *Let V be finite and let $f_1, \dots, f_m \in R$ be a weakly regular sequence of polynomials with zero constant term. Then $m \leq \text{pd}_R R/(f_1, \dots, f_m)$.*

Proof. Let $\mathfrak{p}$ be the kernel of the constant-coefficient map $R \rightarrow k$ (a maximal ideal, and prime because k is a domain). The local ring $R_{\mathfrak{p}}$ is Noetherian, and since localization is flat and all $f_i \in \mathfrak{p}$, the images of $f_1, \dots, f_m$ form a regular sequence of $R_{\mathfrak{p}}$. By Mathlib, the quotient of a Noetherian local ring by a regular sequence of length m has projective dimension exactly m . The localization of the R -module $R/(f)$ at $\mathfrak{p}$ is $R_{\mathfrak{p}}/(f)R_{\mathfrak{p}}$, and localization does not increase projective dimension. Hence $m = \text{pd}_{R_{\mathfrak{p}}} R_{\mathfrak{p}}/(f) \leq \text{pd}_R R/(f)$. $\square$

Theorem 6. *For every field k , every $r \geq 1$ and every m , $\text{pd}_R R/I(H_{r,m}) \geq m$. Consequently:*

- (A) *for every $r \geq 1$ the bound $\text{pd}_R R/I(G) \leq 2r - 1$ fails for the r -partite r -uniform hypergraph $G = H_{r,2r}$ (which has $\text{pd} \geq 2r$);*
- (B) *for every $r \geq 1$ and every N there is an r -partite r -uniform hypergraph with $\text{pd}_R R/I(G) > N$ (take $H_{r,N+1}$), so the supremum over the class is infinite, not $2r - 1$.*

Proof. $I(H_{r,m}) = (f_0, \dots, f_{m-1})$ with $f_i = x^{d_i}$, $d_i = \sum_{j < r} \delta_{(i,j)}$; the d_i have pairwise disjoint supports, and f_i has zero constant term because $r \geq 1$. Apply Lemmas 4 and 5. $\square$

For $r = 2$: four disjoint edges $x_1 y_1, \dots, x_4 y_4$ form a bipartite graph with $\text{pd} \geq 4 > 3$, so the quoted bound “ $\text{pd} \leq 3$ for bipartite graphs” is itself false.

4 Lean formalization

File `lean/Conjecture504/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1, only import Mathlib), namespace `Conjecture504`.

- `Hypergraph`, `Hypergraph.IsUniform`, `Hypergraph.IsPartite`, `edgeIdeal`: the definitions of Section 2.

- `mem_of_monomial_mul_mem` (Lemma 3), `isWeaklyRegular_monomials` (Lemma 4), `length_le_projectiveDimension` (Lemma 5; uses Mathlib's `IsWeaklyRegular.isRegular_of_isLocalization_of_mem`, `ModuleCat.projectiveDimension_quotient_eq_length`, `ModuleCat.projectiveDimension_le_projectiveDimension_of_isLocalizedModule`, `IsLocalizedModule.toLocalizedQuotient'`).
- `matching`, `matching_isUniform`, `matching_isPartite`, `edgeIdeal_matching`, `matching_pd_ge` (the first claim of Theorem 6).
- Main theorems:

```
theorem conjecture504_upper_bound_false (k : Type u) [Field k] (r : Nat)
  (hr : 0 < r) :
  Not (forall (V : Type) [Fintype V] (G : Hypergraph V),
    G.IsUniform r -> G.IsPartite r ->
    projectiveDimension (ModuleCat.of (MvPolynomial V k)
      (MvPolynomial V k / edgeIdeal k G)) <=
      ((2 * r - 1 : Nat) : WithBot ENat))
```

```
theorem conjecture504_unbounded (k : Type u) [Field k] (r : Nat) (hr : 0 < r)
  (N : Nat) :
  exists (V : Type) (_ : Fintype V) (G : Hypergraph V),
    G.IsUniform r /\ G.IsPartite r /\
    (N : WithBot ENat) < projectiveDimension
      (ModuleCat.of (MvPolynomial V k) (MvPolynomial V k / edgeIdeal k G))
```

(ASCII transcription of the Lean source; the Lean quotient symbol is U+29F8, written exttt/above.)

Axiom audit. `lake build` succeeds; `#print axioms` for both main theorems reports exactly `[propext, Classical.choice, Quot.sound]`. No `sorry`, no `native_decide`, no added axioms.